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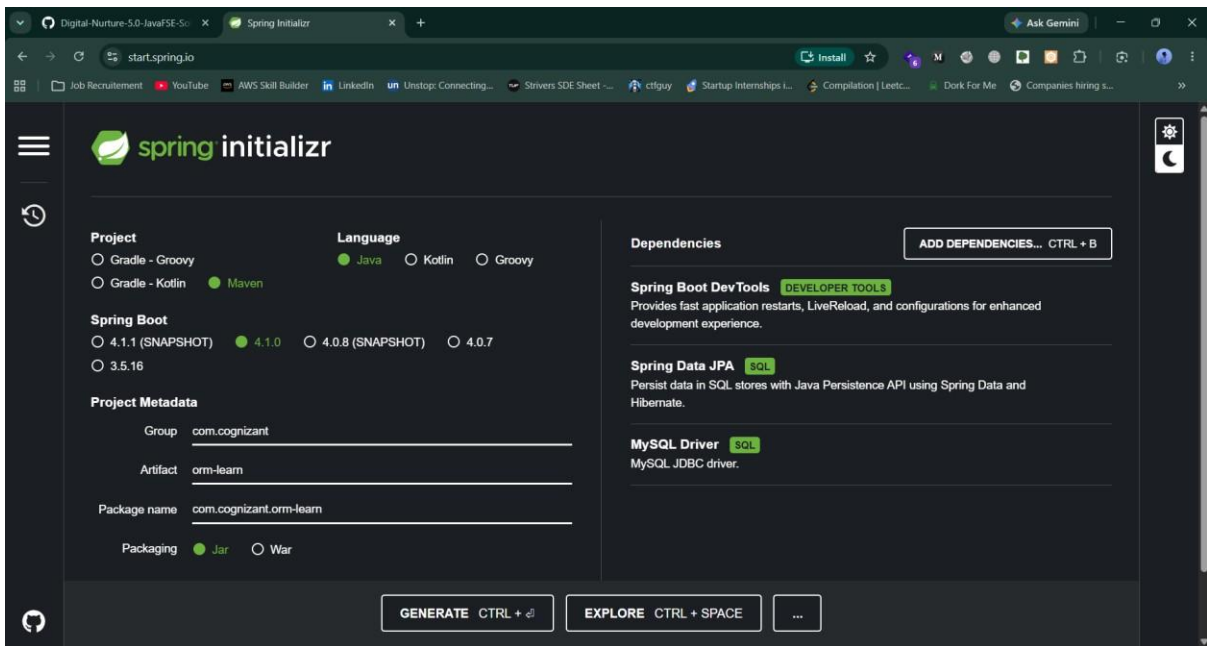
DN-5.0 JavaFSE

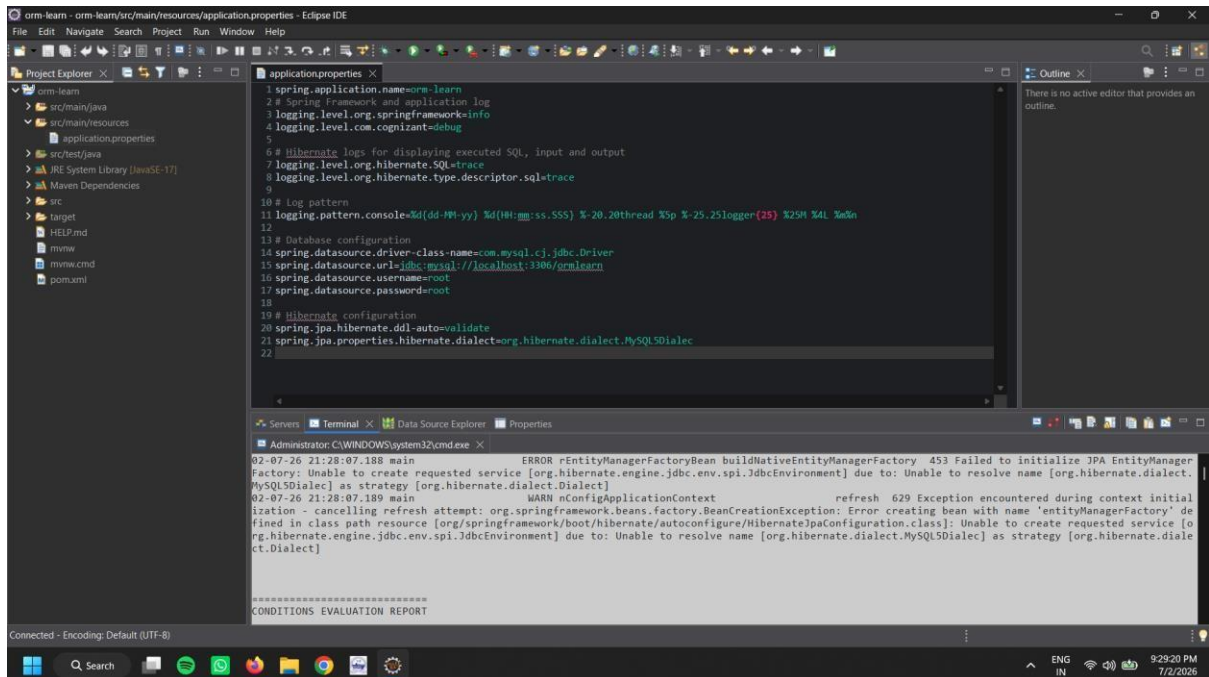
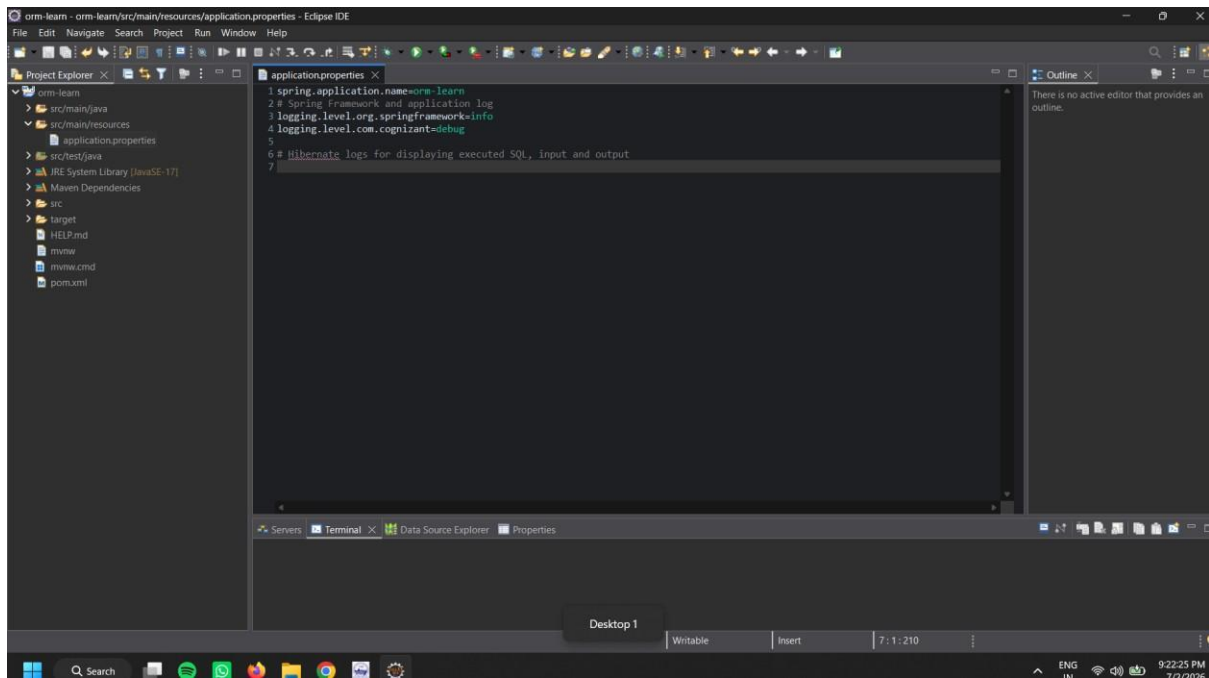
Cognizant-Hands-on-Exercises Java FSE

WEEK-2 : Spring Data JPA with SpringBoot, Hibernate

EXERCISE 1: Spring Data JPA – Quick Example

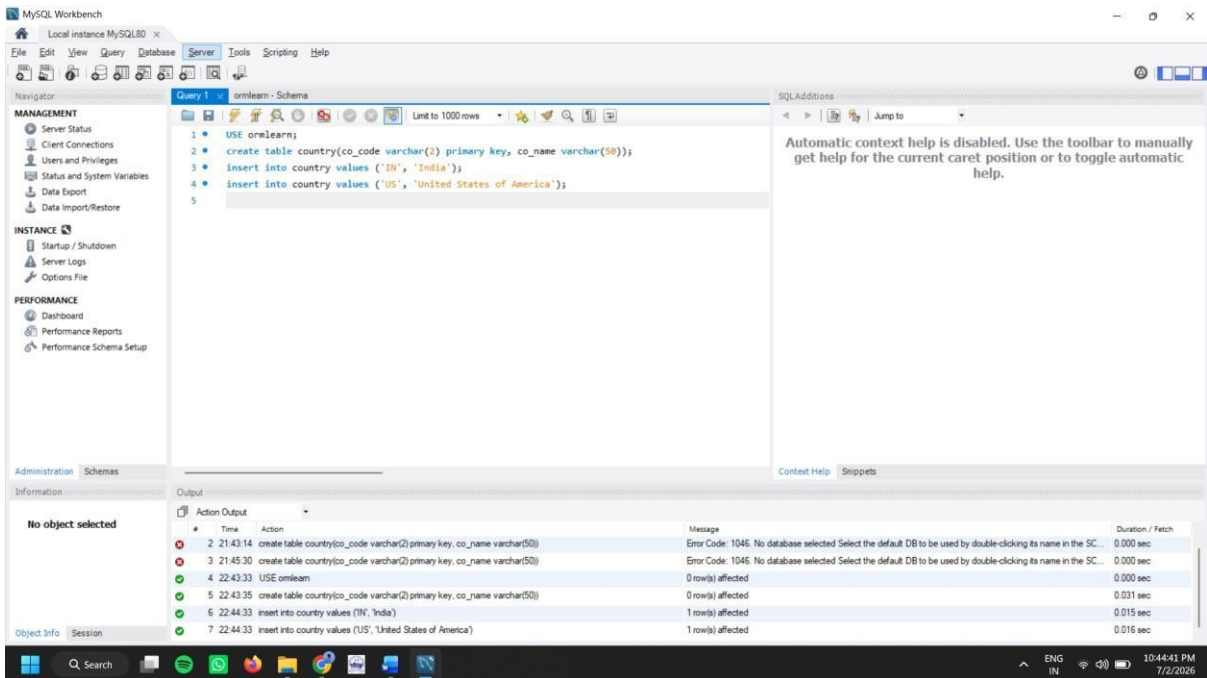
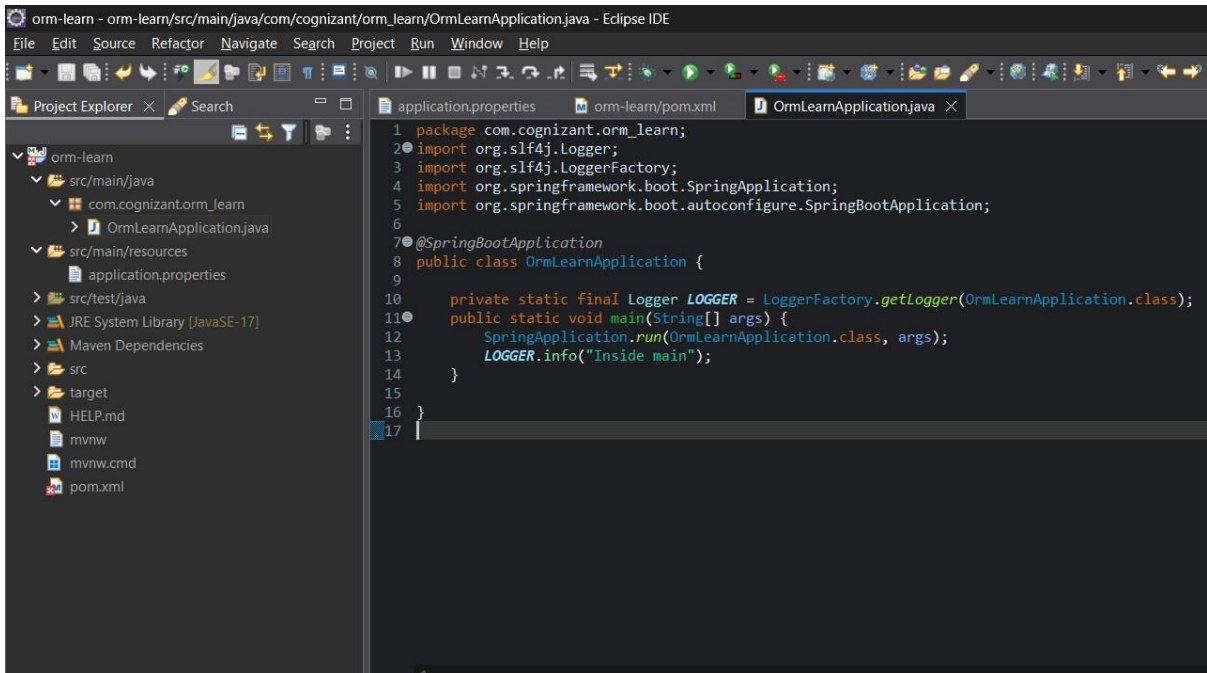
- Go to <https://start.spring.io/>
- Change Group as “com.cognizant”
- Change Artifact Id as “orm-learn”
- In Options > Description enter "Demo project for Spring Data JPA and Hibernate"
- Click on menu and select "Spring Boot DevTools", "Spring Data JPA" and "MySQL Driver"
- Click Generate and download the project as zip
- Extract the zip in root folder to Eclipse Workspace
- Import the project in Eclipse "File > Import > Maven > Existing Maven Projects > Click Browse and select extracted folder > Finish"
- Create a new schema "ormlearn" in MySQL database. Execute the following commands to open MySQL client and create schema.
- In orm-learn Eclipse project, open `src/main/resources/application.properties` and include the below database and log configuration.
- Build the project using '`mvn clean package -Dhttp.proxyHost=proxy.cognizant.com -Dhttp.proxyPort=6050 -Dhttps.proxyHost=proxy.cognizant.com -Dhttps.proxyPort=6050 -Dhttp.proxyUser=123456`' command in command line. Include logs for verifying if `main()` method is called.

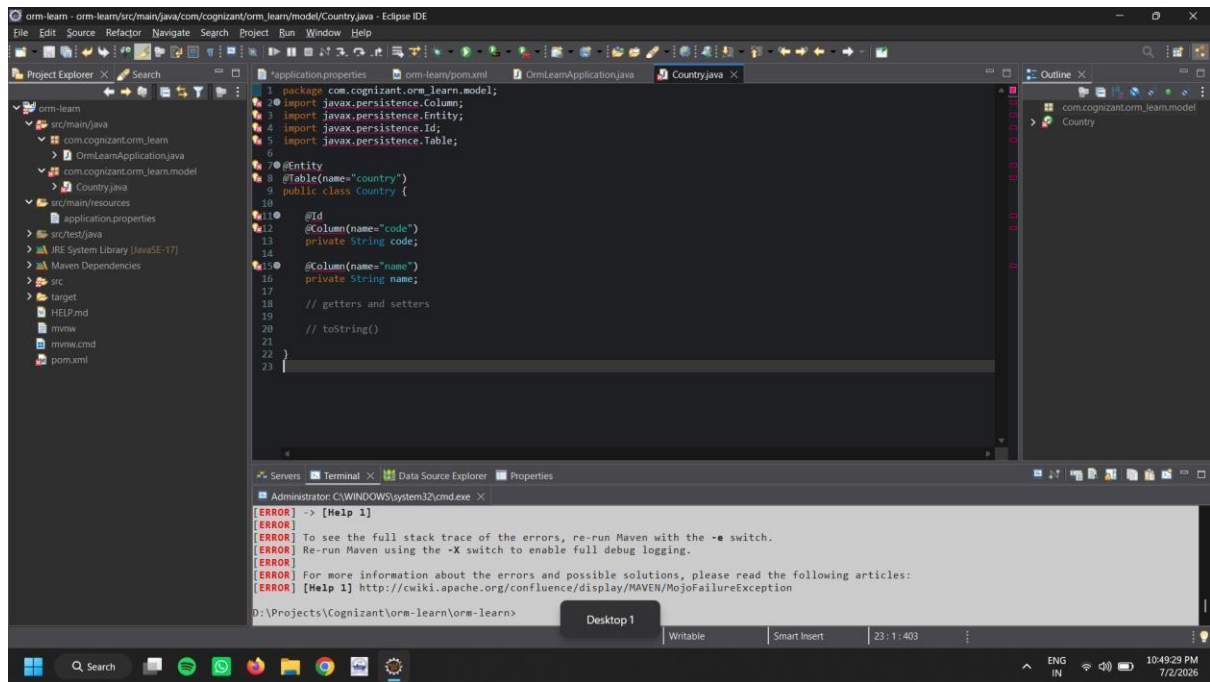




Persistence Class - com.cognizant.orm-learn.model.Country

- Open Eclipse with orm-learn project
- Create new package com.cognizant.orm-learn.model
- Create Country.java, then generate getters, setters and toString() methods.
- Include @Entity and @Table at class level
- Include @Column annotations in each getter method specifying the column name.





EXERCISE 2 Difference between JPA, Hibernate and Spring Data JPA

Java Persistence API (JPA)

- JSR 338 Specification for persisting, reading and managing data from Java objects
- Does not contain concrete implementation of the specification
- Hibernate is one of the implementation of JPA
- ORM Tool that implements JPA
- Spring Data JPA
- Does not have JPA implementation, but reduces boiler plate code
- This is another level of abstraction over JPA implementation provider like Hibernate
- Manages transactions

Refer code snippets below on how the code compares between Hibernate and Spring Data JPA

Hibernate

```
/* Method to CREATE an employee in the database */ public
Integer addEmployee(Employee employee){
    Session session = factory.openSession();
    Transaction tx = null;
    Integer employeeID = null;

    try {
        tx = session.beginTransaction();
        employeeID = (Integer) session.save(employee);    tx.commit();
    } catch (HibernateException e) {
        if (tx != null) tx.rollback();
        e.printStackTrace();    } finally {
        session.close();
    }
    return employeeID;
}
```

Spring Data JPA EmployeeRepository.java *public interface EmployeeRepository extends JpaRepository<Employee, Integer> {*

}

EmployeeService.java *@Autowired private*

EmployeeRepository employeeRepository;

@Transactional public void

addEmployee(Employee employee) {

employeeRepository.save(employee);

}

EXERCISE 3 Implement services for managing Company

An application requires for features to be implemented with regards to country. These features needs to be supported by implementing them as service using Spring Data JPA. ¶ Find a country based on country code

- Add new country
- Update country
- Delete country
- Find list of countries matching a partial country name

Before starting the implementation of the above features, there are few configuration and data population that needs to be incorporated. Please refer each topic below and implement the same.

Explanation for Hibernate table creation configuration

- ¶ Moreover the ddl-auto defines how hibernate behaves if a specific table or column is not present in the database.
- create - drops existing tables data and structure, then creates new tables
 - validate - check if the table and columns exist or not, throws an exception if a matching table or column is not found
 - update - if a table does not exists, it creates a new table; if a column does not exists, it creates a new column
 - create-drop - creates the table, once all operations are completed, the table is dropped

MySQL Workbench

Local instance MySQL80 x Myd@127.0.0.1:3306 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

Query 1 omileam - Schema

Limit to 1000 rows

```
231 insert into country (co_code, co_name) values ("UG", "Uganda");
232 insert into country (co_code, co_name) values ("UA", "Ukraine");
233 insert into country (co_code, co_name) values ("AE", "United Arab Emirates");
234 insert into country (co_code, co_name) values ("GB", "United Kingdom");
235 insert into country (co_code, co_name) values ("US", "United States");
236 insert into country (co_code, co_name) values ("UM", "United States Minor Outlying Islands");
237 insert into country (co_code, co_name) values ("UY", "Uruguay");
238 insert into country (co_code, co_name) values ("UZ", "Uzbekistan");
239 insert into country (co_code, co_name) values ("VU", "Vanuatu");
240 insert into country (co_code, co_name) values ("VE", "Venezuela, Bolivarian Republic of");
241 insert into country (co_code, co_name) values ("VN", "Viet Nam");
242 insert into country (co_code, co_name) values ("VG", "Virgin Islands, British");
243 insert into country (co_code, co_name) values ("VI", "Virgin Islands, U.S.");
244 insert into country (co_code, co_name) values ("WF", "Wallis and Futuna");
245 insert into country (co_code, co_name) values ("EH", "Western Sahara");
246 insert into country (co_code, co_name) values ("YE", "Yemen");
247 insert into country (co_code, co_name) values ("ZM", "Zambia");
248 insert into country (co_code, co_name) values ("ZD", "Zimbabwe");
249 insert into country (co_code, co_name) values ("AX", "Åland Islands");
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

No object selected

Action Output

#	Time	Action	Message	Duration / Fetch
3	21:45:30	create table country(co_code varchar(2) primary key, co_name varchar(50))	Error Code: 1046. No database selected Select the default DB to be used by double-clicking its name in the SC...	0.000 sec
4	22:43:33	USE omileam	0 row(s) affected	0.000 sec
5	22:43:35	create table country(co_code varchar(2) primary key, co_name varchar(50))	0 row(s) affected	0.031 sec
6	22:44:33	insert into country values (IN, "India")	1 row(s) affected	0.015 sec
7	22:44:33	insert into country values (US, "United States of America")	1 row(s) affected	0.016 sec
8	22:57:05	insert into country (co_code, co_name) values ("AF", "Afghanistan")	1 row(s) affected	0.000 sec

Object Info Session

Ready

ENG IN 10:58:29 PM 7/2/2026